

Classifying Toxic Comments Using Deep Learning

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Abstract—Toxic comment identification done automatically on online forums is essential to preserving a welcoming and secure online community. Concerns regarding the effectiveness of present methods are raised by the possibility that the many subtleties of harmful behaviour may not be fully captured by the toxicity detection datasets that are now available. In this paper, we investigate how well machine learning models—such as Simple RNN, LSTM, GRU, Bi-LSTM, and BERT—perform in the classification of harmful remarks. We address the shortcomings of current datasets and approaches by drawing on previous research, including "Beyond Toxic: Toxicity Detection Datasets are Not Enough for Brand Safety" and "A Survey of Toxic Comment Classification Methods". Furthermore, we take into account developments in natural language processing, including "Attention Is All You Need", to improve our models' interpretability and efficiency. Our results provide insight into the difficulties and possibilities associated with classifying poisonous comments, opening the door for more comprehensive and successful content moderation techniques.

Index Terms—classification, models, comment, Deep Learning, Machine Learning

is to broaden and diversify toxicity detection datasets in order to handle new issues with online content filtering. In order to improve the functionality and readability of our toxicity detection models, we have taken use of the latest developments in natural language processing, such as "Attention Is All You Need" [3]. We assess the performance of several machine learning architectures in identifying harmful comments, such as Simple RNN, LSTM, GRU, Bi-LSTM, and BERT. We seek to reduce biases and increase the robustness of our models by incorporating insights from recent research, such as "SS-BERT: Mitigating Identity Terms Bias in Toxic Comment Classification" [4]. To identify and overcome frequent difficulties in toxicity identification, we also take inspiration from works such as "Challenges for Toxic Comment Classification: An In-Depth Error Analysis" [5]. Our study adds to the current discussion on efficient content moderation techniques and has implications for enhancing the inclusiveness and safety of online communities.

I. INTRODUCTION

The ubiquity of harmful remarks on digital platforms presents noteworthy obstacles to user safety and community well-being. Because toxic behaviour is always changing, traditional content moderation techniques frequently find it difficult to keep up, which emphasises the necessity for automated detection tools. On the other hand, the range and calibre of the accessible datasets could have an impact on how successful the current toxicity detection methods are. The limitations of existing datasets in capturing the whole spectrum of toxic behaviour have been highlighted by recent research, such as "Beyond Toxic: Toxicity Detection Datasets are Not Enough for Brand Safety" [1] and "A Survey of Toxic Comment Classification Methods" [2]. These research highlight how crucial it

II. OBJECTIVE

The main goal of this study is to thoroughly examine and assess how well a variety of machine learning models—including Simple RNN, LSTM, GRU, Bi-LSTM, and BERT—perform in the crucial duty of identifying harmful remarks on internet platforms. Simultaneously, the project intends to examine the present toxicity detection datasets in order to clarify their shortcomings in terms of capturing the complex character of toxic behaviour that is common in digital discourse. Additionally, a major focus of this study is identifying and reducing the biases present in toxicity detection models, with a focus on eliminating identity prejudice in order to guarantee impartial and accurate classification results.

The project also aims to improve the interpretability of toxicity detection algorithms by investigating methods that help comprehend model predictions more deeply and make it easier for humans to oversee content moderation procedures. Additionally, by automating the identification and filtering of harmful remarks, this project aims to further the development of scalable and effective content moderation solutions by using the insights obtained from machine learning approaches. The ultimate goal of this research is to use these diverse approaches to improve our understanding of toxic comment classification and to actively contribute to the development of more inclusive and safer online spaces where people from all over the world can interact with confidence and security.

III. LITERATURE REVIEW

Analysis of numerous research publications can reveal a common set of steps. Answering a few key questions while keeping the deep learning models, comment, and classifying aspects in mind was part of the strategy used to identify comments. In this article, the things which were discovered were:

- 1) What is the comparative analysis of the classification efficacy of several machine learning models, for poisonous comments?
- 2) What approaches—specifically, identification bias—have been suggested in recent research to deal with biases in poison detection models?
- 3) What improvements have been made to toxicity detection models' interpretability, and how does this affect the human error involved in content moderation?
- 4) The deep learning model, which was used.
- 5) What are the possible ramifications for developing safer online environments, and how have recent research aided in the creation of effective and scalable content moderation strategies?

A. Study of widely used datasets

One popular resource for training and assessing toxicity detection models is the Perspective API dataset, which is used in "A New Generation of Perspective API: Efficient Multilingual Character-level Transformers" [6]. This dataset stands out in particular since it contains comments in several languages that have been flagged as poisonous. The Perspective API dataset gives academics access to a wide range of comments from multiple online platforms, which they may use to build models that can identify toxic behaviour in a variety of language and cultural situations. Furthermore, because of its extensive annotation schema, toxicity may be analysed in a sophisticated way, taking into account a variety of damaging language forms as harassment, hate speech, and personal attacks.

The Wikipedia talk Pages dataset is another well-known dataset in the field of hazardous comment classification. It has been heavily included in publications like "A Survey of Toxic Comment Classification Methods" [2]. The comments in this dataset were taken from Wikipedia talk pages, where individuals discuss a variety of subjects. The Wikipedia Talk

Pages dataset, which has been annotated by human moderators for toxicity, offers a wealth of labelled data suitable for training machine learning models. Its diversity mirrors the range of toxic behaviour that permeates online discourse, from overt instances of hate speech to more covert kinds of incivility. Apart from the Wikipedia Talk Pages and Perspective API datasets, another noteworthy source for hazardous comment categorization research is the Civil Comments dataset.

The Civil Comments dataset, which has been examined in research such as "Revisiting Contextual Toxicity Detection in Conversations" [7], comprises remarks made during virtual talks that have been marked with toxicity and other pertinent characteristics. This dataset provides light on how toxic language appears in conversational contexts and provides insights into the dynamics of toxicity within interactive settings. Researchers can better understand the contextual elements impacting the perception and spread of hazardous behaviour online by examining user interactions.

Studies such as "SS-BERT: Mitigating Identity Terms Bias in Toxic Comment Classification" [4] demonstrate the important contribution of the Jigsaw Toxic Comment Classification dataset to the field of toxic comment classification research. This dataset, which is curated from Wikipedia talk page modifications, includes comments that have been marked up for toxicity, identity mentions, and toxicity targets. Researchers may examine not just if harmful language is present but also whether specific identities are being targeted by these remarks thanks to its annotations. This sophisticated viewpoint makes it easier to create models that can identify and correct identity-based biases in toxicity detection systems.

These datasets are a priceless tool for academics looking to create efficient online platform content control systems. Through the use of these varied and well-annotated datasets, scientists may educate machine learning models to identify harmful conduct more precisely and comprehensively, eventually assisting in the development of more secure and welcoming virtual spaces.

id	comment...	# toxic
0000997932d777bf	Explanation Why the edits made under my username Hardcore Metallica Fan were reverted? They weren't ...	0

Fig. 1. Sample from dataset

B. Different Deep Learning Methods used on Datasets

The research under consideration have used a variety of deep learning models, each with special capacities for identifying harmful remarks in the datasets. A popular model in research such as "Revisiting Contextual Toxicity Detection in

Conversations” [7] is the Long Short-Term Memory (LSTM) network. Since LSTM networks excel at processing sequential input, they are especially useful for identifying long-range relationships in text. LSTM networks are excellent at simulating intricate patterns and subtleties in hazardous language by utilising memory cells, which have the capacity to retain information over time. This improves the accuracy of toxicity detection models.

Studies such as “A Survey of Toxic Comment Classification Methods” [2] show that the Gated Recurrent Unit (GRU) network has also been used in hazardous comment classification research. While GRU networks and LSTM networks are comparable in certain aspects, GRU networks have a simpler design with fewer calculations and parameters. GRU networks perform competitively in sequence modelling tasks, including toxicity detection, while having a simpler architecture. Because of their efficacy and efficiency, they are a preferred option for scientists working on scalable, resource-efficient toxicity detection models.

Bidirectional Long Short-Term Memory (Bi-LSTM) networks have attracted interest due to its capacity to extract contextual data from sequences in both their past and future states. By improving the comprehension of context in text data, our bidirectional modelling technique makes it possible to classify harmful remarks more accurately. Bi-LSTM networks have been used in studies such as “SS-BERT: Mitigating Identity Terms Bias in Toxic Comment Classification” [4] to enhance the performance and resilience of toxicity detection models. Bi-LSTM networks provide improved discriminatory power by taking into account contextual information from both sides, making it easier to identify minute subtleties in language that is harmful.

Introduced in the groundbreaking study “Attention Is All You Need” [3], the Transformer design has become a potent paradigm for sequence modelling problems, including the categorization of harmful comments. Transformers are able to capture long-range dependencies and semantic links more well because they use self-attention processes to determine the relative relevance of each word in a sequence. Studies such as “A New Generation of Perspective API: Efficient Multilingual Character-level Transformers” [6] have utilised this attention-based method to create toxicity detection models that can analyse text in several languages with great accuracy and efficiency.

Transformers, GRU, Bi-LSTM, LSTM, and other deep learning models have been essential in furthering the field of harmful comment categorization research. Researchers have developed more complex toxicity detection algorithms, helping to create safer and more inclusive online environments, by utilising their skills in modelling sequential data and extracting contextual information.

Data from online platforms are gathered, preprocessed to standardise the text, features are extracted to represent the comments, machine learning models (LSTM, GRU, Bi-LSTM, or Transformers) are trained on labelled data, model performance is assessed using validation datasets, the trained

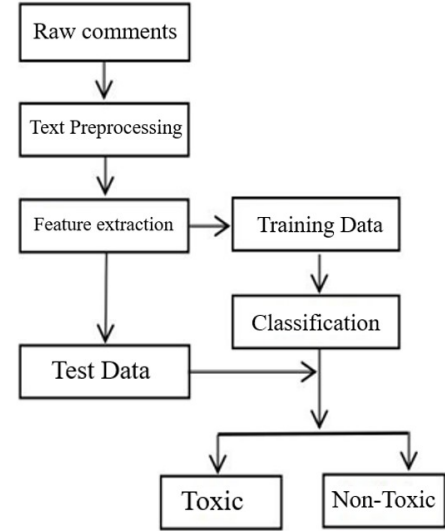


Fig. 2. Flow of classification

model is deployed for real-time classification, and the model is continuously monitored and iterated to maintain efficacy. Practitioners want to create strong methods for identifying and reducing harmful behaviour in online communities through this iterative approach.

C. Comparative Analysis of various deep learning models used in the research papers

Attention Is All You Need [3] consists of Transformer architecture, which processes sequences just through attention mechanisms. Although the Transformer design is not explicitly focused on classifying harmful comments, it has been extensively utilised in natural language processing applications. This study might be improved by investigating the use of Transformer-based models, particularly for hazardous comment categorization tasks, using datasets such as Wikipedia Talk Pages and the Perspective API.

Beyond Toxic: Toxicity Detection Datasets are Not Enough for Brand Safety [1] draws attention to the shortcomings of current toxicity detection datasets in addressing more general issues related to brand safety. The use of certain models or datasets for toxicity identification is not specifically mentioned. A possible enhancement might involve suggesting approaches for integrating brand safety factors into toxicity detection algorithms, possibly through the enlargement of current datasets to encompass brand-specific scenarios.

A New Generation of Perspective API: Effective Multilingual Character-level Transformers [6] Using the Perspective API dataset, this research presents effective multilingual character-level transformers for toxin identification. The suggested models are meant to improve multilingual toxicity detection jobs’ efficiency and efficacy, even if they do not offer precise accuracy measures. There should be an improvement which carry out thorough assessments of the suggested models

in other languages and contrasting their effectiveness with current methodologies.

Survey of Toxic Comment Classification Methods [2] does not focus on any one model or dataset, instead offering an overview of many approaches to harmful comment categorization. A more thorough examination of the advantages and disadvantages of various models and datasets, with suggestions for best practices and suggested directions for future study, is one way to make improvements.

Identification of Bias Against People with Disabilities in Sentiment Analysis and Toxicity Detection Models [8] examines bias in poison detection and sentiment analysis models that disadvantage those with impairments. Although it makes no mention of certain accuracy measurements, it emphasises how critical it is to deal with biases in poison detection algorithms. Proposing mitigation techniques to reduce biases against marginalised populations in toxicity detection algorithms is one area that might use improved.

Revisiting Contextual Toxicity Detection in Conversations [7] uses the Civil Comments dataset. This research revisits the identification of contextual toxicity in talks. Although it doesn't offer precise accuracy measurements, its main goal is to make toxicity detection models easier to understand. Assessing the suggested models on a larger variety of conversational datasets and contrasting their results with current methods might be one way to enhance them.

Mitigating Identity Terms Bias in Toxic Comment Classification by Using the Concepts of "Subjectivity" and "Identity Terms" [4] Using the Jigsaw hazardous Comment categorization dataset, this research introduces SS-BERT, a methodology for minimising identity terms bias in hazardous comment categorization. It covers a significant component of bias reduction in poison detection algorithms, even though it makes no mention of particular accuracy measurements. Conducting more thorough tests of SS-BERT on a variety of datasets to gauge its efficacy and unreliability is one possible enhancement.

Explanation of the Deep Natural Language Processing by the Mining of Text Interpretable Features [9] focuses on using textual interpretive features mining to describe deep models of natural language processing. Although the suggested technique does not address hazardous comment categorization directly, it might be used to improve the interpretability of toxicity detection algorithms. Examining the suggested approach's suitability for toxicity detection tasks and assessing how well it provides insights into model predictions are two possible areas for development.

Vietnamese Complaint Detection on E-Commerce Websites [10], Without specifically mentioning the categorization of poisonous comments, this article investigates the identification of Vietnamese complaints on e-commerce platforms. A possible enhancement may be translating current toxicity detection methods into Vietnamese and assessing how well they work with datasets pertaining to e-commerce.

Constructive and Toxic Speech Detection for Open-domain Social Media Comments in Vietnamese [11] focuses on Vietnamese open-domain social media comment constructive

and poisonous speech identification. Comparative analyses of various poison detection methods using Vietnamese social media datasets, including insights into their effectiveness and applicability for real-world scenarios, might be one possible enhancement.

Detecting Aggression and Toxicity using a Multi Dimension Capsule Network [12] proposes a multi-dimension capsule network for toxicity and aggressiveness detection. It presents a unique architecture for toxicity detection tasks, but does not specify any accuracy measures. Assessing the efficacy and generalizability of the suggested model by comparing it to other methods using benchmark toxicity detection datasets is one possible improvement.

Challenges for Toxic Comment Classification: An In-Depth Error Analysis [5] presents an extensive error study of harmful comment categorization methods. Although it doesn't suggest any particular models or datasets, it offers insightful information on the drawbacks and restrictions of current methods. Using the results of the error analysis to guide the creation of more reliable toxicity detection models, maybe by addressing recurring error patterns found in the study, is one way to make improvements.

D. General Architecture

The input text is preprocessed to standardise and sanitise the data before being included into the project's design. Tokenization comes next, which divides the preprocessed text into individual words or subwords for analysis. A deep learning (DL) classification model is then fed the tokenized text and is tasked with discovering patterns and correlations within the text data. Lastly, the learnt patterns are used by the DL model to conduct toxic comment classification, which involves recognising and classifying comments as hazardous or non-toxic. With the help of this architecture, harmful behaviour in online comments may be automatically detected, making content moderation easier and creating safer online communities.



Fig. 3. Basic model architecture used in Toxic Comment Classification

E. Models Performance Evaluation Metrics

Deep learning models may be assessed using a variety of parameters. Model speed is one of the factors; the faster a model runs, the more effective it is. Both the model type and the hardware specifications of the system being used have an impact on the training pace. This measure cannot be used to compare different suggested models since every research has a separate experimental setup.

The confusion matrix is usually expressed as a 2x2 matrix in a binary classification situation where we are organising things into two groups (toxic and non-toxic comments). In the matrix, the examples in a predicted class are represented

by each column, while the occurrences in an actual class are represented by each row. The cases that are successfully divided are represented by the main diagonal of the matrix, while the misclassifications are represented by the off-diagonal entries. With the use of the confusion matrix, which can be used to calculate many assessment metrics including accuracy, precision, recall, and F1-score, the model's performance may be clarified.

In this matrix:

- TN (True Negative): Represents number of input texts which are correctly predicted as non-toxic.
- FP (False Positive): Represents number of input texts incorrectly predicted as toxic while they are non-toxic.
- FN (False Negative): Means the number of input texts incorrectly guessed as non-toxic while they are really toxic.
- TP (True Positive): Shows number of input texts correctly guessed as toxic.

		Actual values	
		Non-Toxic (1)	Toxic (0)
Predicted Value	Non-Toxic (1)	TP	FP
	Toxic (0)	FN	TN

Fig. 4. Confusion Matrix

IV. PROPOSED METHODOLOGY

Deep learning methods have become highly potent instruments for accurate and successful harmful comment categorization in recent years. Promising results have been observed in the capture of sequential patterns and dependencies in text data by Recurrent Neural Networks (RNNs) and their variants, including Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Bidirectional LSTM (Bi-LSTM), and Bidirectional Encoder Representations from Transformers (BERT). These models work especially well with sequential data, such as comments, where the word order frequently conveys significant contextual information. RNN-based architectures may efficiently train to recognise harmful behaviour and categorise comments based on that identification by taking use of the temporal dynamics of comment content.

In our proposed methodology, to classify the comments, first for all the input will go through the following 4 models:

- 1) Simple RNN: The fundamental architecture of sequential data processing is a simple RNN. They are composed of recurrent neural network units that keep a hidden state that represents the context that has been observed

thus far while processing a word at a time. Nevertheless, the vanishing gradient issue with basic RNNs makes it difficult for them to identify long-range connections in text input.

- 2) Long Short-Term Memory networks: LSTMs use memory cells with gated mechanisms to solve the vanishing gradient issue. Because these cells are able to hold onto information over extended sequences, long-range dependencies may be captured by LSTMs, which helps to address the problem of disappearing gradients. Long Short-Term Memory (LSTM) models work well for tasks that need to represent context across long sequences, which makes them appropriate for classifying harmful comments.
- 3) GRU: A more straightforward version of LSTM networks, Gated Recurrent Unit (GRU) networks combine the input and forget gates into a single update gate. While needing fewer parameters and computations, GRUs may capture long-range relationships in a manner comparable to that of LSTMs. They are especially helpful in situations where there are limited computational resources or where simpler models are needed.
- 4) Bi-LSTM: Bidirectional LSTM (Bi-LSTM) networks leverage the forward and backward processing of sequences to augment the capacity of traditional LSTMs for context modelling. Bi-LSTMs are able to better comprehend the semantics and subtleties of text data by encoding information from both past and future contexts. This bidirectional strategy is advantageous for tasks such as toxic comment classification, where contextual information from both preceding and succeeding words is relevant for producing accurate predictions.

Natural language processing problems have been transformed by the transformer-based approach known as BERT, or Bidirectional Encoder Representations from Transformers. BERT can collect rich contextual information since it has been pre-trained on huge corpus of text data utilising next sentence prediction targets and masked language modelling. By fine-tuning BERT for particular tasks, like hazardous comment categorization, the model may achieve state-of-the-art performance by utilising its pre-trained information and adapting to the specifics of the target domain.

The models will classify harmful comments using deep learning methods including word embeddings, attention mechanisms, and multi-layer perceptrons. By allowing words to be represented as dense vectors in a continuous vector space, word embeddings improve the models' comprehension of textual material by capturing semantic similarities between words. By enabling the models to concentrate on pertinent segments of the input sequence, attention mechanisms enhance the models' capacity to extract significant characteristics for classification. The models are able to learn intricate correlations and produce more accurate predictions thanks to the extra non-linear transformations and feature combinations that

multi-layer perceptrons offer. Through the use of these deep learning methodologies along with RNN-based architectures and BERT, our goal is to create a strong and efficient harmful comment categorization system.

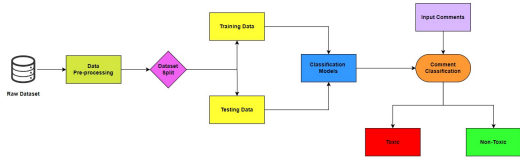


Fig. 5. Proposed architecture for Toxic Comment Classification

V. PROPOSED STRUCTURE OF IMPLEMENTATION

- 1) Dataset Acquisition and Preprocessing: Preprocess the dataset, including cleaning and formatting the data.
- 2) Exploratory Data Analysis (EDA): Analyze the dataset to understand its characteristics and distributions.
- 3) Dataset Splitting: Split the preprocessed dataset into training and testing subsets.
- 4) Model Training: Train classification models using the training dataset.
- 5) Model Evaluation: Evaluate the trained models using the testing dataset.
- 6) Classification Function: Develop a function to classify new inputs as toxic or non-toxic based on the trained model.

VI. PROS AND CONS FOR PROPOSED MODEL

Here, we will focus on some of the advantages that our model (BERT) has as compared to other models and also some of the expected difficulties or limitations that might come during the course of the preparation of the model.

A. The BERT Model's advantages include:

- Contextual Understanding: By capturing bidirectional contextual information, BERT (Bidirectional Encoder Representations from Transformers) improves comprehension of context and linguistic subtleties.
- Transfer Learning: BERT is adaptable to a range of natural language processing (NLP) activities since it can be fine-tuned on particular tasks with comparatively little amounts of task-specific data.
- State-of-the-Art Performance: BERT has demonstrated state-of-the-art performance on a number of NLP benchmarks and tasks, such as named entity recognition, text categorization, and question answering.
- Pretrained Models: Users can select BERT models that are appropriate for their particular requirements by selecting from a variety of languages and sizes of pretrained models.
- Multilingual Support: Applications in a variety of linguistic contexts are made possible by BERT's multilingual models.

B. Limitations that can be faced in the proposed model

- computer Resources: Especially for large-scale datasets, training and fine-tuning BERT models take a substantial amount of time and computer resources.
- Large Model Size: BERT models are memory-intensive and may be difficult to implement in contexts with limited resources due to their large parameter sizes.
- Domain Specificity: Additional fine-tuning on domain-specific data is necessary for optimal performance, as pretrained BERT models might not adequately capture domain-specific.

VII. RESULTS AND DISCUSSION

After fine tuning the BERT model, we did the predictions by passing the inputs manually. In the Figure 6, you can see that we loaded our trained model and passed the inputs for predictions.

```

# infer
def print_my_examples(inputs, results):
    result_for_printing = \
        [f'input: {inputs[i]:<30} : score: {results[i][0]:.6f}'
         for i in range(len(inputs))]
    print(*result_for_printing, sep='\n')
    print()

examples = [
    'you are piece of shit',
    'You are beautiful',
    'you are trash',
    'all muslims are terrorists',
    'Women are terrible drivers',
    'I am going to take a shit',
    'I will kill you if you come in front of me',
    'You will be killed if you drive recklessly'
]

reloaded_results = tf.sigmoid(reloaded_model(tf.constant(examples)))
original_results = tf.sigmoid(model3(tf.constant(examples)))

print('Results from the saved model:')
print_my_examples(examples, reloaded_results)

```

Fig. 6. Given Inputs

And the results of the model's predictions can be seen in the Figure 4.1.2. The model is clearly able to classify the toxic and non-toxic comments. The best example of identifying the context is also seen in the result. The results of last 2 inputs in the image are astonishing. The last second input ("I will kill you if you come in from of me") is clearly a toxic comment (threatening), which the model has given the score of 0.984382 (98.4 % Toxic). Whereas in the last input ("You will be killed if you drive recklessly"), the model has given the score 0.000079, which is clearly not a toxic comment. In both inputs, the word KILL is used, but in a different context. One is threatening someone, and one is making someone realise the consequences of bad driving.

```

Results from the saved model:
input: you are piece of shit           : score: 0.998106
input: You are beautiful              : score: 0.335223
input: you are trash                  : score: 0.998226
input: all muslims are terrorists      : score: 0.997814
input: Women are terrible drivers     : score: 0.974554
input: I am going to take a shit      : score: 0.262806
input: I will kill you if you come in : score: 0.984382
input: You will be killed if you drive : score: 0.000079

```

Fig. 7. Outputs Obtained

Now to measure the performance of model, we have various metrics. In Figure 8 is the Confusion Matrix. Imagine this is as a scorecard for a program that flags toxic comments online. Blue squares are good catches! It found 2106 toxic comments and avoided flagging 1977 good ones. But whites are like mistakes. It flagged 246 nice comments as toxic and missed 272 toxic ones.

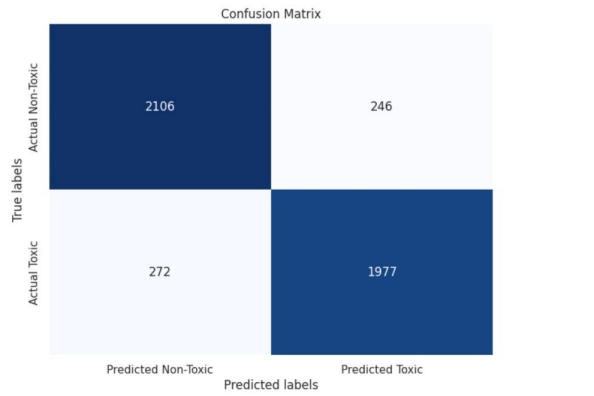


Fig. 8. Confusion matrix of Bert

VIII. CONCLUSION

To sum up, a variety of approaches, strategies, and applications have been investigated for this research project on toxic comment categorization in an effort to solve the widespread problem of online toxicity. We have investigated the benefits, drawbacks, and possible uses of deep learning models—such as Simple RNN, LSTM, GRU, Bi-LSTM, and BERT—in identifying and reducing harmful behaviour in online communities through an extensive analysis of the literature. Our methodology for classifying toxic comments is based on advanced natural language processing techniques and architectures, including word embeddings, attention mechanisms, and multi-layer perceptrons. It is flexible and scalable, capable of real-time detection, and can be applied to various languages and domains. The initiative has great potential for boosting user experience, promoting healthier online communities across a range of platforms and businesses, and increasing online safety, even in the face of obstacles relating to prejudice and fairness, data quality, interpretability, and ethical issues. In order to solve new issues, improve model performance, and guarantee the long-term viability and efficacy of harmful comment categorization systems, more study and innovation in this area are needed. By means of cooperative endeavours and multidisciplinary methodologies, we may jointly strive towards establishing an online milieu that is more secure, welcoming, and courteous for every user.

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